(d) After scaling: $45/8 \approx 5.7$. Without scaling, logits of magnitude ~ 45 push softmax into saturation (near one-hot), making gradients vanish and the model unable to attend to multiple positions simultaneously.

Mathematics for Machine Learning — Chapter 2

Pencil-and-Paper Exercises

Abstract

Who this is for. Someone working through MML Chapter 2 (Linear Algebra, §2.1–2.3 and §2.7) alongside mechanistic interpretability study. These exercises cover systems of equations, matrix properties (rank, transpose, inverse), and linear mappings — the foundations that make transformer weight matrices readable.

How to use this. Work through each section in order. Do the exercises by hand before checking the solutions starting on page 9.

Contents

1 Systems of Linear Equations **2**

2 Matrix Properties **3**

3 Linear Mappings **5**

4 Norms, Inner Products, and Orthogonality **7**

Solutions **9**

1 Systems of Linear Equations

Why this matters. A system of linear equations $A\mathbf{x} = \mathbf{b}$ is the fundamental operation underlying almost everything in a transformer. The question "what input $\mathbf{x}$ produces output $\mathbf{b}$ through matrix A?" is exactly what you are asking when you reverse-engineer a weight matrix. Gaussian elimination — the row-reduction method for solving systems — also builds intuition for rank and linear dependence, which reappear constantly in circuit analysis.

Exercise 1.1 — Solving a 2×2 system

Solve the following system using substitution or elimination:

$$2x + y = 5, \qquad x - y = 1$$

(a) Write this as an augmented matrix $[A \mid \mathbf{b}]$.

(b) Apply row operations to reach row echelon form.

(c) Read off the solution.

Exercise 1.2 — Gaussian elimination on a 3×3 system

Solve:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \\ 3 \end{bmatrix}$$

(a) Write the augmented matrix $[A \mid \mathbf{b}]$.

(b) Eliminate x_1 from rows 2 and 3 using row operations.

(c) Eliminate x_2 from row 3.

(d) Back-substitute to find x_3, then x_2, then x_1.

Exercise 1.3 — Counting solutions

For each augmented matrix below (already in row echelon form), state whether the system has **no solution**, **one solution**, or **infinitely many solutions**, and give a brief reason.

(a) $\begin{bmatrix} 1 & 2 & | & 3 \\ 0 & 1 & | & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 & | & 3 \\ 0 & 0 & | & 5 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 2 & 3 & | & 6 \\ 0 & 1 & 1 & | & 3 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$

Exercise 1.4 — Homogeneous systems

A *homogeneous* system has the form $A\mathbf{x} = \mathbf{0}$.

(a) Why does a homogeneous system always have at least one solution?

(b) Find the general solution to:

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Section 4: Norms, Inner Products, and Orthogonality

Solution 4.1 — Computing norms

(a) $\|\mathbf{v}\|_2 = \sqrt{9+16} = \sqrt{25} = 5$.

(b) $\|\mathbf{u}\|_2 = \sqrt{1+1+1+1} = 2$.

(c) $\frac{1}{\sqrt{64}} = \frac{1}{8} = 0.125$.

(d) $\|\mathbf{v}\|_1 = 3 + 4 = 7$. Larger than $\|\mathbf{v}\|_2 = 5$. In general $\|\mathbf{v}\|_1 \geq \|\mathbf{v}\|_2$.

Solution 4.2 — Inner products

(a) $\langle \mathbf{v}, \mathbf{v} \rangle = 4 + 9 = 13 > 0$. ✓

(b) $\mathbf{u}^\top A\mathbf{v} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2$.

(c) $f(\mathbf{v}, \mathbf{v}) = 0 \cdot 0 - 1 \cdot 1 = -1 < 0$. Not positive definite — not a valid inner product.

Solution 4.3 — Orthogonality and projection

(a) $\mathbf{u} \cdot \mathbf{v} = 2 + 0 - 2 = 0$. Yes, orthogonal.

(b) $\text{proj}_{\mathbf{a}}\mathbf{b} = \frac{3}{1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$. This is the shadow of $\mathbf{b}$ onto the x-axis — the x-component of $\mathbf{b}$.

(c) $\hat{\mathbf{a}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. $\hat{\mathbf{a}} \cdot \mathbf{b} = \frac{1}{\sqrt{2}}(3+4) = \frac{7}{\sqrt{2}}$. Projection: $\frac{7}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{7}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3.5 \\ 3.5 \end{bmatrix}$.

(d) Yes — the logit lens and direct path analysis are both inner product operations.

Solution 4.4 — Orthogonal complements

(a) Need $w_1 + w_2 = 0$. Two independent solutions: $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

(b) $\dim(U) = 1$, $\dim(U^\perp) = 2$, sum $= 3 = n$. General rule: $\dim(U) + \dim(U^\perp) = n$.

(c) Project onto $\hat{\mathbf{a}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$: coefficient $= \frac{1}{\sqrt{2}}(2+3+0) = \frac{5}{\sqrt{2}}$. Component along $\mathbf{a}$: $\frac{5}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 2.5 \\ 0 \end{bmatrix}$. Remainder: $\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 2.5 \\ 2.5 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.5 \\ 1 \end{bmatrix}$.

(d) Those components are completely invisible to the query projection — they contribute nothing to attention scores and are effectively discarded by the head.

Solution 4.5 — Angles and the $\sqrt{d}$ scaling

(a) $\cos\theta = \frac{0+0}{\sqrt{1}\sqrt{2}} \cdot \ldots$ Actually: $\mathbf{u} \cdot \mathbf{v} = 1$, $\|\mathbf{u}\| = 1$, $\|\mathbf{v}\| = \sqrt{2}$. $\cos\theta = \frac{1}{\sqrt{2}}$, so $\theta = 45°$.

(b) $\mathbf{q} \cdot \mathbf{k} = \cos(45°) = \frac{1}{\sqrt{2}} \approx 0.707$.

(c) Each has expected norm 8; $\mathbf{q} \cdot \mathbf{k} \approx \|\mathbf{q}\|\|\mathbf{k}\| \cos\theta = 8 \cdot 8 \cdot \frac{1}{\sqrt{2}} \approx 45$.

(b) Yes. An explicit bijection: $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$. This is linear, injective, and surjective.

(c) $\mathbb{R}^{768}$ and $\mathbb{R}^{64}$ are not isomorphic (different dimensions). W_Q maps from the larger space to the smaller one — it is a surjective (onto) but not injective map. Its kernel has dimension $768 - 64 = 704$: a 704-dimensional subspace of the residual stream maps to zero and is completely invisible to the head.

Solution 3.6 — Change of basis

(a) $\mathbf{v} = 3\mathbf{e}_1 + (-2)\mathbf{e}_2$.

(b) Solve $c_1 + c_2 = 3$, $c_1 - c_2 = -2$. Adding: $2c_1 = 1$, so $c_1 = \frac{1}{2}$, $c_2 = \frac{5}{2}$.

(c) Projecting a residual stream vector onto a specific direction (e.g., the "John" direction in the IOI circuit) is exactly expressing that vector's coordinates in a new basis where that direction is one of the basis vectors. The projection coefficient tells you "how much" of that concept is present.

(c) The solution set you found is a *subspace*. What is its dimension? What does that tell you about the null space of this matrix?

Exercise 1.5 — Geometric interpretation of solutions

Consider a system $A\mathbf{x} = \mathbf{b}$ where A is 2×2 and the two rows of the augmented matrix represent two lines in the plane.

(a) Sketch (roughly) three scenarios: (i) the lines intersect at a point, (ii) the lines are parallel but distinct, (iii) the lines are identical. Match each to: unique solution, no solution, infinitely many solutions.

(b) Solve the system $x + y = 3$, $2x + 2y = 6$. Which case is this? Describe the solution set geometrically.

(c) Solve $x + y = 3$, $2x + 2y = 7$. Which case is this?

Exercise 1.6 — Back-substitution practice

The following matrix is already in row echelon form. Back-substitute to find the solution:

$$\left[\begin{array}{ccc|c} 2 & -1 & 3 & 7 \\ 0 & 3 & -1 & 4 \\ 0 & 0 & 2 & 6 \end{array}\right]$$

Show each step explicitly.

2 Matrix Properties

Why this matters. Transformer weight matrices have specific structural properties — the QK matrix is low-rank (at most `d_head` rank), the attention pattern matrix is square, and composition of $W_V W_O$ is a rank-`d_head` matrix in $d_{\text{model}} \times d_{\text{model}}$ space. The concepts in this section — rank, transpose, identity, inverse — are what let you read those claims in papers without getting lost.

Exercise 2.1 — Rank by row reduction

Find the rank of each matrix by reducing it to row echelon form.

(a) $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{bmatrix}$

(b) $B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

(c) $C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$ (a 3×2 matrix — what is the maximum possible rank?)

Exercise 2.2 — Rank and the weight matrices

In GPT-2 Small, each attention head has $d_{\text{model}} = 768$ and $d_{\text{head}} = 64$. The query weight matrix W_Q has shape 768×64.

(a) What is the maximum possible rank of W_Q?

(b) The QK matrix is defined as $W_{QK} = W_Q W_K^\top$. What shape is W_{QK}?

(c) What is the maximum possible rank of W_{QK}?

(d) Explain in plain language what it means for W_{QK} to have rank 64 in a 768-dimensional space. What are the heads "blind" to?

Exercise 2.3 — Transpose properties

Let:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

(a) Compute $A^\top$.

(b) Compute $AA^\top$ and $A^\top A$. What are their shapes?

(c) Is $AA^\top$ symmetric? (A matrix M is symmetric if $M = M^\top$.) Is $A^\top A$ symmetric? State a general rule.

(d) In attention, the raw scores are $QK^\top$. If Q has shape $(n \times d_h)$ and K has shape $(n \times d_h)$, what shape is $QK^\top$? Why is this the right shape for an attention pattern?

Exercise 2.4 — The identity and inverse

(a) Write down the 3×3 identity matrix I_3.

(b) Verify that $I_3\mathbf{v} = \mathbf{v}$ for $\mathbf{v} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$.

(c) Find the inverse of $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ using the formula for 2×2 matrices: $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

(d) Verify your answer: compute AA^{-1} and confirm you get I.

(e) What is the determinant of $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$? Does this matrix have an inverse? Why or why not?

Exercise 2.5 — Symmetric and orthogonal matrices

(a) Which of the following are symmetric?

$$P = \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 0 & 3 \\ 2 & 3 & 5 \end{bmatrix}$$

(b) A matrix U is *orthogonal* if $U^\top U = I$. Verify that $U = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ is orthogonal.

(c) If U is orthogonal, what is U^{-1}? (You do not need to compute — just read it from the definition.)

(d) Orthogonal matrices preserve vector lengths: $\|U\mathbf{v}\| = \|\mathbf{v}\|$. Verify this for $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and the matrix U above.

Exercise 2.6 — Matrix multiplication order matters

Section 3: Linear Mappings

Solution 3.1 — Verifying linearity

(a) **Linear.** $f(\mathbf{u}+\mathbf{v}) = 3(u_1+v_1) - 2(u_2+v_2) = f(\mathbf{u}) + f(\mathbf{v})$; $f(\alpha\mathbf{v}) = \alpha f(\mathbf{v})$. Both hold.

(b) **Not linear.** $f(2\mathbf{v}) = 2v_1 \cdot 2v_2 = 4v_1v_2 \neq 2f(\mathbf{v})$.

(c) **Not linear.** $f(\mathbf{0}) = 0 + 1 = 1 \neq 0$. Linear maps must send $\mathbf{0}$ to $\mathbf{0}$.

(d) **Linear.** Matrix multiplication satisfies both linearity conditions for any fixed A.

Solution 3.2 — Kernel (null space)

(a) Row reduce: $R_2 \leftarrow R_2 - 2R_1$: $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$. Free variables: $x_2 = s$, $x_3 = t$. Then $x_1 = -2s - 3t$.

$\ker(A) = \text{span}\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$

(b) Nullity $= 2$.

(c) $\text{rank}(A) = 1$ (one nonzero row). $1 + 2 = 3 =$ number of columns. ✓

(d) A large kernel means the query projection discards a large subspace of the residual stream — many residual stream directions contribute nothing to the head's attention pattern.

Solution 3.3 — Image (column space)

(a) Shape: 3×2. Domain: $\mathbb{R}^2$, codomain: $\mathbb{R}^3$.

(b) Rank 2 (columns are independent).

(c) Column vectors: $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

(d) For $\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$: need $x_1 = 2$, $x_2 = 3$, check third component $2 + 3 = 5$. ✓In the image. For $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$: need $x_1 = 1$, $x_2 = 0$, check $1 + 0 = 1 \neq 2$. **Not** in the image.

Solution 3.4 — Composition of linear maps

(a) $C = BA = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 3 & 3 \end{bmatrix}$. Shape: 2×3.

(b) $f(\mathbf{v}) = \begin{bmatrix} 1+1 \\ 2+1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$. $g(f(\mathbf{v})) = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$.

(c) $C\mathbf{v} = \begin{bmatrix} 2+0+2 \\ 0+6+3 \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}$. ✓

(d) W_V is $(d_h \times d_m)$, W_O is $(d_m \times d_h)$... Actually per MML convention with row vectors: W_V maps $(d_m \to d_h)$ and W_O maps $(d_h \to d_m)$, so $W_V W_O$ is $(d_m \times d_m)$. Maximum rank: $d_h = 64$.

Solution 3.5 — Isomorphism and dimension

(a) No. A bijective linear map preserves dimension, but $\mathbb{R}^2$ has dimension 2 and $\mathbb{R}^3$ has dimension 3. No such bijection exists.

(a) P and R are symmetric ($P^\top = P$, $R^\top = R$). Q is not: $Q^\top = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \neq Q$.

(b) $U^\top U = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = I$. ✓

(c) $U^{-1} = U^\top$ (read directly from $U^\top U = I$).

(d) $U\mathbf{v} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. $\|U\mathbf{v}\| = \frac{1}{\sqrt{2}}\sqrt{1^2 + 1^2} = \frac{\sqrt{2}}{\sqrt{2}} = 1 = \|\mathbf{v}\|$. ✓

Solution 2.6 — Matrix multiplication order matters

(a) $AB = \begin{bmatrix} 7 & 2 \\ 3 & 1 \end{bmatrix}$, $BA = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$. Not equal.

(b) The QK matrix $W_Q W_K^\top$ defines the attention pattern; $W_K^\top W_Q$ has a different shape and different meaning entirely. Order encodes the direction of the mapping.

(c) $(AB)^\top = \begin{bmatrix} 7 & 3 \\ 2 & 1 \end{bmatrix}$. $B^\top A^\top = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 2 & 1 \end{bmatrix}$. Equal. ✓

Solution 2.7 — Trace and the identity

(a) $\mathrm{tr}(I_3) = 1 + 1 + 1 = 3$.

(b) $\mathrm{tr}(A) = 3 + (-1) + 5 = 7$.

(c) $AB = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$, $\mathrm{tr}(AB) = 5$. $BA = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$, $\mathrm{tr}(BA) = 5$. ✓

(a) Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$. Compute AB and BA. Are they equal?

(b) This is the non-commutativity of matrix multiplication: in general $AB \neq BA$. Give one concrete example of why the order of W_Q, W_K matters in transformers.

(c) Compute $(AB)^\top$ and $B^\top A^\top$. Are they equal? This is the *transpose of a product* rule: $(AB)^\top = B^\top A^\top$.

Exercise 2.7 — Trace and the identity

The *trace* of a square matrix A is the sum of its diagonal entries: $\mathrm{tr}(A) = \sum_i A_{ii}$.

(a) Compute $\mathrm{tr}(I_3)$.

(b) Compute $\mathrm{tr}(A)$ for $A = \begin{bmatrix} 3 & 1 & 0 \\ 2 & -1 & 4 \\ 0 & 2 & 5 \end{bmatrix}$.

(c) It can be shown that $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ even though $AB \neq BA$ in general. Verify this for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

3 Linear Mappings

Why this matters. MML §2.7 frames matrices not as arrays of numbers but as *functions between vector spaces*. This is the right way to think about transformer weight matrices: W_Q is a function that maps residual stream vectors to query space; W_E (the embedding matrix) is a function that maps token indices to vectors. The kernel and image of these functions tell you what information they preserve, what they discard, and what they can express.

Exercise 3.1 — Verifying linearity

A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is *linear* if: (i) $f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$, and (ii) $f(\alpha\mathbf{v}) = \alpha f(\mathbf{v})$ for any scalar α.

For each function below, determine whether it is linear. If not, give a counterexample.

(a) $f(x_1, x_2) = 3x_1 - 2x_2$

(b) $f(x_1, x_2) = x_1 x_2$

(c) $f(x_1, x_2) = x_1 + 1$

(d) $f(\mathbf{x}) = A\mathbf{x}$ for any fixed matrix A

Exercise 3.2 — Kernel (null space)

The *kernel* of a linear map f is the set of all inputs that map to $\mathbf{0}$: $\ker(f) = \{\mathbf{x} : f(\mathbf{x}) = \mathbf{0}\}$.

For $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$:

(a) Solve $A\mathbf{x} = \mathbf{0}$ to find $\ker(A)$.

(b) What is the dimension of $\ker(A)$? (This is the *nullity* of A.)

(c) The rank-nullity theorem states: $\mathrm{rank}(A) + \mathrm{nullity}(A) = n$, where n is the number of columns. Verify this holds for A.

(d) In terms of transformer circuits: if $\ker(W_Q)$ is large, what does that mean about how much information the query projection "throws away"?

Exercise 3.3 — Image (column space)

The *image* of a linear map A is the set of all vectors A can produce: $\mathrm{im}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}$.

For $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$:

(a) What is the shape of A? What are its domain and codomain?

(b) What is $\mathrm{rank}(A)$?

(c) The image of A is spanned by the columns of A. Write out the two column vectors.

(d) Is the vector $\begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$ in $\mathrm{im}(A)$? Is $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$? (Check whether each can be written as $A\mathbf{x}$ for some $\mathbf{x}$.)

Exercise 3.4 — Composition of linear maps

Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ be given by matrix A, and $g : \mathbb{R}^2 \to \mathbb{R}^2$ be given by matrix B:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

(a) Compute $C = BA$ (the matrix for $g \circ f$). What shape is C?

(b) Apply f then g to $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ step by step.

(c) Apply C directly to $\mathbf{v}$ and confirm you get the same answer.

(d) In a transformer, the OV circuit $W_V W_O$ is a composition of two linear maps. W_V maps from d_{model} to d_{head}, and W_O maps from d_{head} back to d_{model}. What is the shape of $W_V W_O$? What is its maximum rank?

Exercise 3.5 — Isomorphism and dimension

Two vector spaces are *isomorphic* if there is a bijective linear map between them.

(a) Are $\mathbb{R}^2$ and $\mathbb{R}^3$ isomorphic? Why or why not? (Hint: think about what a bijective linear map would have to do to dimensions.)

(b) The space of 2×2 matrices $\mathbb{R}^{2\times 2}$ has dimension 4. Is it isomorphic to $\mathbb{R}^4$? Describe an explicit bijection (a way to turn any 2×2 matrix into a 4-dimensional vector).

(c) In a transformer, the residual stream lives in $\mathbb{R}^{d_{\text{model}}}$. The query space for one head lives in $\mathbb{R}^{d_{\text{head}}}$. Are these isomorphic? What does W_Q do to bridge them, and why is information necessarily lost?

Exercise 3.6 — Change of basis (conceptual)

(a) The standard basis for $\mathbb{R}^2$ is $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Write $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ as a linear combination of $\mathbf{e}_1$ and $\mathbf{e}_2$.

Section 2: Matrix Properties

Solution 2.1 — Rank by row reduction

(a) $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - R_1$: $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & -2 & -2 \end{bmatrix}$, swap $R_2 \leftrightarrow R_3$: two nonzero rows. $\mathrm{rank}(A) = 2$.

(b) $R_3 \leftarrow R_3 - R_1 - R_2$: $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. $\mathrm{rank}(B) = 2$.

(c) Maximum possible rank is $\min(3, 2) = 2$. After reduction: two independent rows. $\mathrm{rank}(C) = 2$.

Solution 2.2 — Rank and the weight matrices

(a) $\min(768, 64) = 64$.

(b) W_Q is (768×64), $W_K^\top$ is (64×768), so W_{QK} is (768×768).

(c) At most 64 (bounded by the intermediate dimension).

(d) The head lives in a 64-dimensional subspace of a 768-dimensional space. Any residual stream direction orthogonal to this subspace is invisible to the head's attention pattern — it contributes nothing to the query or key computation.

Solution 2.3 — Transpose properties

(a) $A^\top = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

(b) $AA^\top$ is $(2 \times 3)(3 \times 2) = (2 \times 2)$; $A^\top A$ is $(3 \times 2)(2 \times 3) = (3 \times 3)$.

(c) Both are symmetric. General rule: $AA^\top$ and $A^\top A$ are always symmetric for any A.

(d) Q is $(n \times d_h)$, $K^\top$ is $(d_h \times n)$, so $QK^\top$ is $(n \times n)$ — one score per (query position, key position) pair, which is exactly an attention weight matrix.

Solution 2.4 — The identity and inverse

(a) $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(b) $I_3\mathbf{v} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ (unchanged).

(c) $\det(A) = 2 \cdot 3 - 1 \cdot 5 = 1$. So $A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$.

(d) $AA^{-1} = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. ✓

(e) $\det = 2 \cdot 2 - 4 \cdot 1 = 0$. No inverse — the matrix is *singular* (rank 1, not full rank).

Solution 2.5 — Symmetric and orthogonal matrices

(b) $2x + 2y = 6$ is $2 \times (x + y = 3)$ — the same line. Infinitely many solutions: $\{(t, 3 - t) : t \in \mathbb{R}\}$. A whole line of solutions.

(c) Row reduce: $R_2 - 2R_1$ gives $0 = 1$ — no solution. Parallel lines.

Solution 1.6 — Back-substitution practice

From row 3: $2x_3 = 6 \Rightarrow x_3 = 3$. From row 2: $3x_2 - 3 = 4 \Rightarrow x_2 = \frac{7}{3}$. From row 1: $2x_1 - \frac{7}{3} + 9 = 7 \Rightarrow 2x_1 = 7 + \frac{7}{3} - 9 = \frac{1}{3} \Rightarrow x_1 = \frac{1}{6}$.

(b) Now suppose you want to express $\mathbf{v}$ in a new basis: $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Find scalars c_1, c_2 such that $c_1\mathbf{b}_1 + c_2\mathbf{b}_2 = \mathbf{v}$.

(c) In mechanistic interpretability, researchers often project residual stream activations onto specific directions (e.g., the "unembedding direction" of a token, or the top singular vectors of a weight matrix) to understand what information is present. How does this relate to the idea of expressing a vector in a different basis?

4 Norms, Inner Products, and Orthogonality

Why this matters. MML Chapter 3 (§3.1–3.4) generalises the dot product to the concept of an *inner product*, and introduces norms and orthogonality rigorously. In transformers: the $\sqrt{d_{\text{head}}}$ scaling in attention is a direct consequence of norm growth with dimension; orthogonality explains why features in different subspaces don't interfere; and projection onto a direction is an inner product operation. These exercises build the geometric intuition behind those mechanisms.

Exercise 4.1 — Computing norms

The ℓ^2 (Euclidean) norm of a vector $\mathbf{v} \in \mathbb{R}^n$ is $\|\mathbf{v}\|_2 = \sqrt{\sum_i v_i^2}$.

(a) Compute $\|\mathbf{v}\|_2$ for $\mathbf{v} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$.

(b) Compute $\|\mathbf{u}\|_2$ for $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$.

(c) If each component of $\mathbf{u}$ is drawn from $\mathcal{N}(0, 1)$ with d components, the expected ℓ^2 norm grows as $\sqrt{d}$. This is why attention logits are scaled by $\frac{1}{\sqrt{d_{\text{head}}}}$. For $d_{\text{head}} = 64$, what is the scaling factor?

(d) Compute the ℓ^1 norm $\|\mathbf{v}\|_1 = \sum_i |v_i|$ for $\mathbf{v} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$. Is it larger or smaller than $\|\mathbf{v}\|_2$?

Exercise 4.2 — Inner products

The standard dot product is one example of an inner product. A general inner product $\langle \cdot, \cdot \rangle$ must be: (i) symmetric, (ii) linear in the first argument, (iii) positive definite ($\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, with equality only if $\mathbf{v} = \mathbf{0}$).

(a) Verify that the standard dot product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v}$ satisfies positive definiteness for $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$.

(b) A *weighted* inner product uses a positive definite matrix A: $\langle \mathbf{u}, \mathbf{v} \rangle_A = \mathbf{u}^\top A \mathbf{v}$. Compute $\langle \mathbf{u}, \mathbf{v} \rangle_A$ for $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$.

(c) Is $f(\mathbf{u}, \mathbf{v}) = u_1 v_1 - u_2 v_2$ a valid inner product on $\mathbb{R}^2$? Check positive definiteness by trying $\mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Exercise 4.3 — Orthogonality and projection

Two vectors are *orthogonal* if their inner product is zero: $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

(a) Are $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$ orthogonal?

(b) The projection of $\mathbf{b}$ onto $\mathbf{a}$ is: $\text{proj}_{\mathbf{a}}\mathbf{b} = \frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}\mathbf{a}$ Project $\mathbf{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ onto $\mathbf{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Interpret the result geometrically.

(c) Compute the projection of $\mathbf{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ onto $\mathbf{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (normalise $\mathbf{a}$ first).

(d) In mech interp, researchers project residual stream activations onto a "token direction" (e.g., the row of the unembedding matrix for token "John"). This projection is an inner product. A high value means the residual stream is "pointing toward" that token's output direction. Does this feel concrete now?

Exercise 4.4 — Orthogonal complements

The *orthogonal complement* of a subspace U in $\mathbb{R}^n$ is $U^\perp = \{\mathbf{w} : \langle \mathbf{w}, \mathbf{u} \rangle = 0 \text{ for all } \mathbf{u} \in U\}$.

(a) Let $U = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$ in $\mathbb{R}^3$. Find two vectors in $U^\perp$ (i.e., two independent vectors orthogonal to $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$).

(b) What is $\dim(U) + \dim(U^\perp)$? What is the ambient dimension n? State the general rule.

(c) Every vector $\mathbf{x} \in \mathbb{R}^n$ can be uniquely decomposed as $\mathbf{x} = \mathbf{u} + \mathbf{u}^\perp$ where $\mathbf{u} \in U$ and $\mathbf{u}^\perp \in U^\perp$. Decompose $\mathbf{x} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ into its component along $\mathbf{a} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and its orthogonal remainder.

(d) Why does orthogonal decomposition matter for low-rank matrices? If W_Q has rank 64, what happens to residual stream components in $(\text{im}\, W_Q^\top)^\perp$?

Exercise 4.5 — Angles and the $\sqrt{d}$ scaling

The angle θ between two vectors satisfies $\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$.

(a) Find the angle between $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (your answer will be in radians or degrees).

(b) Suppose query vector $\mathbf{q}$ and key vector $\mathbf{k}$ both have unit norm and are at 45° to each other. What is $\mathbf{q} \cdot \mathbf{k}$?

(c) Now suppose $d_{\text{head}} = 64$ and both $\mathbf{q}, \mathbf{k}$ have components drawn from $\mathcal{N}(0, 1)$, so their expected norm is $\sqrt{64} = 8$. What is the expected scale of $\mathbf{q} \cdot \mathbf{k}$ if the angle is still 45°?

(d) After dividing by $\sqrt{d_{\text{head}}} = 8$, what is the scaled attention logit? Explain in one sentence why this scaling is necessary for softmax to produce well-spread attention weights rather than near-one-hot distributions.

Solutions

Section 1: Systems of Linear Equations

Solution 1.1 — 2×2 system

(a) Augmented matrix: $\begin{bmatrix} 2 & 1 & | & 5 \\ 1 & -1 & | & 1 \end{bmatrix}$

(b) $R_1 \leftrightarrow R_2$: $\begin{bmatrix} 1 & -1 & | & 1 \\ 2 & 1 & | & 5 \end{bmatrix}$, then $R_2 \leftarrow R_2 - 2R_1$: $\begin{bmatrix} 1 & -1 & | & 1 \\ 0 & 3 & | & 3 \end{bmatrix}$

(c) $y = 1$, back-substitute: $x - 1 = 1$, so $x = 2$. Solution: $(x, y) = (2, 1)$.

Solution 1.2 — Gaussian elimination on a 3×3 system

(a) $\begin{bmatrix} 1 & 2 & 1 & | & 6 \\ 2 & 1 & 3 & | & 9 \\ 0 & 1 & 1 & | & 3 \end{bmatrix}$

(b) $R_2 \leftarrow R_2 - 2R_1$: $\begin{bmatrix} 1 & 2 & 1 & | & 6 \\ 0 & -3 & 1 & | & -3 \\ 0 & 1 & 1 & | & 3 \end{bmatrix}$

(c) $R_3 \leftarrow R_3 + \frac{1}{3}R_2$: $\begin{bmatrix} 1 & 2 & 1 & | & 6 \\ 0 & -3 & 1 & | & -3 \\ 0 & 0 & \frac{4}{3} & | & 2 \end{bmatrix}$

(d) $x_3 = 2 \cdot \frac{3}{4} = \frac{3}{2}$. From row 2: $-3x_2 + \frac{3}{2} = -3$, so $x_2 = \frac{3}{2}$. From row 1: $x_1 + 3 + \frac{3}{2} = 6$, so $x_1 = \frac{3}{2}$.

Solution 1.3 — Counting solutions

(a) **One solution.** Two equations, two unknowns, no contradiction. Back-substitute: $x_2 = 1$, $x_1 = 1$.

(b) **No solution.** Row 2 says $0 = 5$, which is a contradiction.

(c) **Infinitely many solutions.** The last row is all zeros — the system is underdetermined. One free variable (x_3); solutions form a 1-dimensional family.

Solution 1.4 — Homogeneous systems

(a) $\mathbf{x} = \mathbf{0}$ is always a solution (the *trivial solution*).

(b) Row reduce: $\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ Row 2 is redundant. Free variables: $x_2 = s$, $x_3 = t$. Then $x_1 = -2s + t$. General solution: $\begin{bmatrix} -2s + t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

(c) Dimension 2. The null space is spanned by two independent vectors — the matrix "throws away" a 2-dimensional subspace of its input.

Solution 1.5 — Geometric interpretation of solutions

(a) (i) Lines intersect $\to$ unique solution. (ii) Parallel, distinct $\to$ no solution. (iii) Same line $\to$ infinitely many solutions.